Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to outline the software requirements for the Web-based Order Management System (OMS) developed for **ABC Power Equipment Inc.** This document serves as a guide for the development team, QA engineers, and project stakeholders by providing detailed functional and non-functional requirements, technical stack specifications, and system architectural details.

1.2 Document Conventions

This document is based on standard IEEE SRS guidelines and follows Agile user story mapping (Epics, User Stories).

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Project Managers and Scrum Masters
- Development Team (Front-end, Back-end, Database, DevOps)
- Quality Assurance (QA) Engineers
- Client Stakeholders (ABC Power Equipment Inc.)

1.4 Product Scope

The **Web-based Order Management System (OMS)** (Project Code: W-OMS01) aims to streamline and automate order processing, bill of materials (BOM) generation, purchase order creation, invoicing, and payment tracking. The primary goal is to reduce order processing time (from 5 to 2 days), improve invoice accuracy to 98%, and enhance the on-time delivery rate to 95%.

1.5 References

- Database Design Document (`db_design.pdf`)
- Epic Document (`epic_document.pdf`)

- Project Backlog (`project_backlog.pdf`)
- Project Guidelines (`project_guidline_oms.pdf`)

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2. Overall Description

2.1 Product Perspective

The OMS will be a brand new web-based application replacing manual and time-consuming order processing practices at ABC Power Equipment Inc. The system will feature a centralized platform with role-based access control, allowing various stakeholders (Sales Representatives, Production Managers, Purchasing Officers, Finance Managers, Business Analysts, and Administrators) to interact safely and efficiently.

2.2 Product Functions

The OMS consists of the following core modules (Epics):

1. **Order Management**: Create, edit, and track sales orders.
2. **BOM Management**: Generate detailed Bill of Materials based on product configurations.
3. **Purchase Order Management**: Create, track, and manage purchase orders for required components.
4. **Invoice and Payment Management**: Generate invoices, track payments, and reconcile finances.
5. **Reporting and Analytics**: Access comprehensive reports and dashboards on performance and inventory.
6. **User Management and Security**: Manage role-based access controls and ensure data security.

2.3 User Classes and Characteristics

- **Sales Representatives & Customers**: Utilize the Order Management module to book and track sales orders.
- **Production Managers & Engineers**: Handle the BOM module to plan and manage production accurately.
- **Purchasing Officers & Suppliers**: Use the Purchase Order Management module for procurement.
- **Finance Managers**: Generate invoices and track payments via the Invoice and Payment Management module.

- **Business Analysts & Managers**: Analyze operational trends via the Reporting and Analytics dashboards.
- **Administrators**: Manage user roles, access, and security compliance.

2.4 Operating Environment

- **Cloud Infrastructure**: AWS or Azure (Cloud-based deployment)
- **Front-end Environment**: Modern web browsers (Chrome, Firefox, Safari, Edge) supporting React JS, HTML, CSS, JavaScript. (Note: Project is currently implemented in Angular based on previous repository contexts).
- **Back-end Environment**: Node.js runtime with Express.js.
- **Database Server**: Oracle SQL / SQL Server.

2.5 Design and Implementation Constraints

- The system must integrate seamlessly with Agile practices (CI/CD via Jenkins, containerization with Docker/Kubernetes).
- Must adhere strictly to role-based access control (RBAC) to restrict unauthorized access to financial data.
- The project timeline enforces completion within an 18-week (4.5 months) window.

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3. System Features (Functional Requirements)

3.1 Order Management (Epic 1)

- **Description:** Allow sales representatives and customers to efficiently create, edit, manage, and process sales orders.
- **Key Features:**
  - Create and edit sales orders.
  - Manage order status and track shipments.
  - Generate order confirmations and notifications.
- **Database Entities Impacted:** `Sales_Order` , `Sales_Order_Item` , `Customer` , `Product` .

3.2 BOM Management (Epic 2)

- **Description:** Provide production managers with tools to generate detailed Bill of Materials (BOM).
- **Key Features:**

- Create and maintain product configurations.
- Generate detailed BOMs from sales orders.
- Track changes in BOM structure.

- **Database Entities Impacted:** `Product` , `Product_BOM` .

3.3 Purchase Order Management (Epic 3)

- **Description:** Enable purchasing officers to create and manage purchase orders to replenish required components.
- **Key Features:**
  - Create, submit, and track purchase orders.
  - Manage supplier information and communication.
- **Database Entities Impacted:** `Purchase_Order` , `Purchase_Order_Item` , `Supplier` .

3.4 Invoice and Payment Management (Epic 4)

- **Description:** Help finance managers generate invoices and track incoming payments against orders.
- **Key Features:**
  - Generate and send accurate invoices for sales orders.
  - Track payment status and set up due date alerts.
  - Reconcile payments and invoices.
- **Database Entities Impacted:** `Invoice` , `Payment` .

3.5 Reporting and Analytics (Epic 5)

- **Description:** Supply business analysts and managers with comprehensive operational reports.
- **Key Features:**
  - Dashboards for inventory levels, order statuses, and alerts.
  - Trend and financial performance reporting.

3.6 User Management and Security (Epic 6)

- **Description:** Secure the system against unauthorized access and assign appropriate permissions.
- **Key Features:**
  - Role-based Access Control (RBAC).
  - Centralized user creation, updating, and suspension.

    - Secure authentication protocols.
- **Database Entities Impacted:** `User`.

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4. Non-Functional Requirements (NFR)

4.1 Performance Requirements

- **Order Processing Time:** Maximum average of 2 days from order placement to shipment.
- **System Uptime:** The system must guarantee 99.9% availability.
- **Scalability:** The infrastructure should utilize containerization (Docker, Kubernetes) to easily scale horizontally during peak demand.

4.2 Quality Assurance Attributes

- **Invoice Accuracy:** The system shall maintain an invoice generation accuracy of at least 98%.
- **On-Time Delivery Rate:** Target delivery success rate of 95% supported by real-time tracking.
- **Testing Standard:** Unit tests will be implemented via Jest, with overall code coverage expectations to prevent regression bugs.

4.3 Security Requirements

- All user passwords and sensitive information must be encrypted.
- Data transferred over the web must use HTTPS protocols.
- Access to specific endpoints (e.g., Reports, Finance) will be strictly guarded by server-side role verifications.

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5. Technology Stack Summary

- **Front-end:** Angular, HTML5, Vanilla CSS, JavaScript
- **Back-end:** Springboot
- **Database:** Postgres SQL
- **DevOps & Infrastructure:** Github Actions, Jenkins (CI/CD), Docker, Kubernetes, Ansible
- **Code Quality & Review:** Git, GitHub, SonarQube
- **Project Management:** JIRA, Slack, Microsoft Teams

6. Project Deliverables and Schedule

The total project duration is mapped to 18 weeks, delivered incrementally:

- **Order Management:** 4 weeks
- **BOM Management:** 3 weeks
- **Purchase Order Management:** 4 weeks
- **Invoice and Payment Management:** 3 weeks
- **Reporting & Analytics / User Management:** 4 weeks combined

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This SRS is formulated iteratively based on Agile methodologies and represents the requirements defined in the supporting documents.